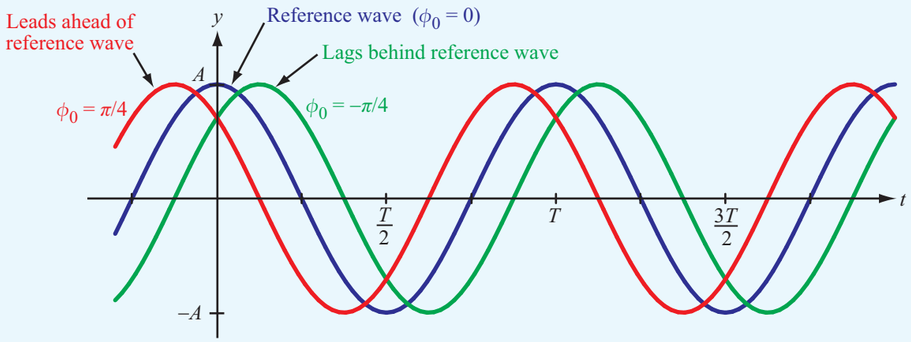


Figure 1-14 Plots of $y(0,t) = A\cos[(2\pi t/T) + \phi_0]$ for three different values of the reference phase ϕ_0.

proportional to ϕ_0. This is illustrated by the plots shown in **Fig. 1-14**. We observe that when ϕ_0 is positive, $y(t)$ reaches its peak value, or any other specified value, sooner than when $\phi_0 = 0$. Thus, the wave with $\phi_0 = \pi/4$ is said to ***lead*** the wave with $\phi_0 = 0$ by a ***phase lead*** of $\pi/4$; and similarly, the wave with $\phi_0 = -\pi/4$ is said to ***lag*** the wave with $\phi_0 = 0$ by a ***phase lag*** of $\pi/4$. A wave function with a negative ϕ_0 takes longer to reach a given value of $y(t)$, such as its peak, than the zero-phase reference function.

▶ When its value is positive, ϕ_0 signifies a phase lead in time, and when it is negative, it signifies a phase lag. ◀

Exercise 1-3: Consider the red wave shown in **Fig. E1.3**. What is the wave's (a) amplitude, (b) wavelength, and (c) frequency given that its phase velocity is 6 m/s?

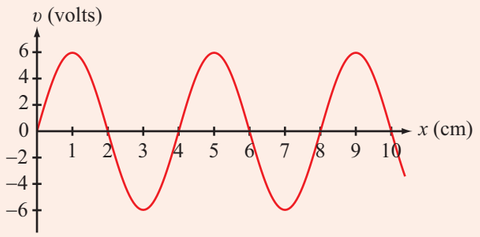


Figure E1.3

Answer: (a) $A = 6$ V, (b) $\lambda = 4$ cm, (c) $f = 150$ Hz. (See EM.)

Exercise 1-4: The wave shown in red in **Fig. E1.4** is given by $v = 5\cos 2\pi t/8$. Of the following four equations:

(1) $v = 5\cos(2\pi t/8 - \pi/4)$,

(2) $v = 5\cos(2\pi t/8 + \pi/4)$,

(3) $v = -5\cos(2\pi t/8 - \pi/4)$,

(4) $v = 5\sin 2\pi t/8$,

(a) which equation applies to the green wave? (b) which equation applies to the blue wave?

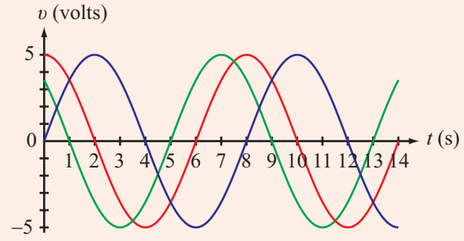


Figure E1.4

Answer: (a) #2, (b) #4. (See EM.)

相位超前、滞后与波形练习

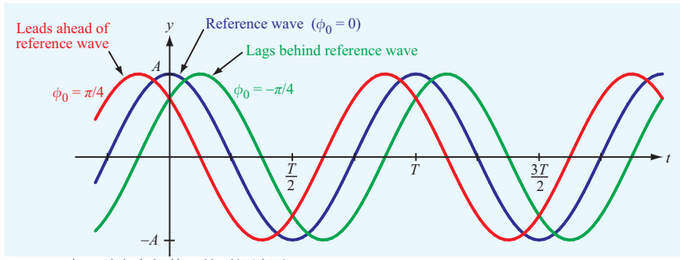


图1-14 $y(0,t) = A\cos(2\pi t/T + \varphi_0)$在三种参考相位下的时间波形。

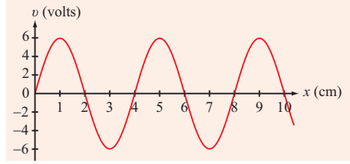
（承上页）在固定位置，$\varphi_0 > 0$的波比$\varphi_0 = 0$的参考波更早达到同一相位状态。例如$\varphi_0 = \pi/4$表示相位超前$\pi/4$；$\varphi_0 = -\pi/4$表示相位滞后$\pi/4$。相对于零相位参考波，负参考相位需要更晚的时刻才能达到相应波峰。

图中Leads ahead：超前；Lags behind：滞后；Reference wave：参考波。此处“超前 / 滞后”针对固定位置、采用$\cos(\omega t + \varphi_0)$及$\omega > 0$的时间约定；不能脱离约定只看相位常数正负。

练习1-3

图E1.3给出红色波形，已知相速度为6 m/s，求（a）振幅；（b）波长；（c）频率。

答案： $A = 6$ V，$\lambda = 4$ cm $= 0.04$ m，$f = u_p/\lambda = 6/0.04 = 150$ Hz。



图E1.3 横轴x（cm），纵轴电压v（V）。

练习1-4

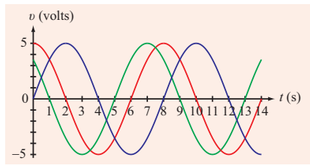
图E1.4中的红色波形为$v = 5\cos(2\pi t/8)$。以下哪个表达式分别对应（a）绿色波形；（b）蓝色波形？

(1) $v = 5\cos(2\pi t/8 - \pi/4)$,

(2) $v = 5\cos(2\pi t/8 + \pi/4)$,

(3) $v = -5\cos(2\pi t/8 - \pi/4)$,

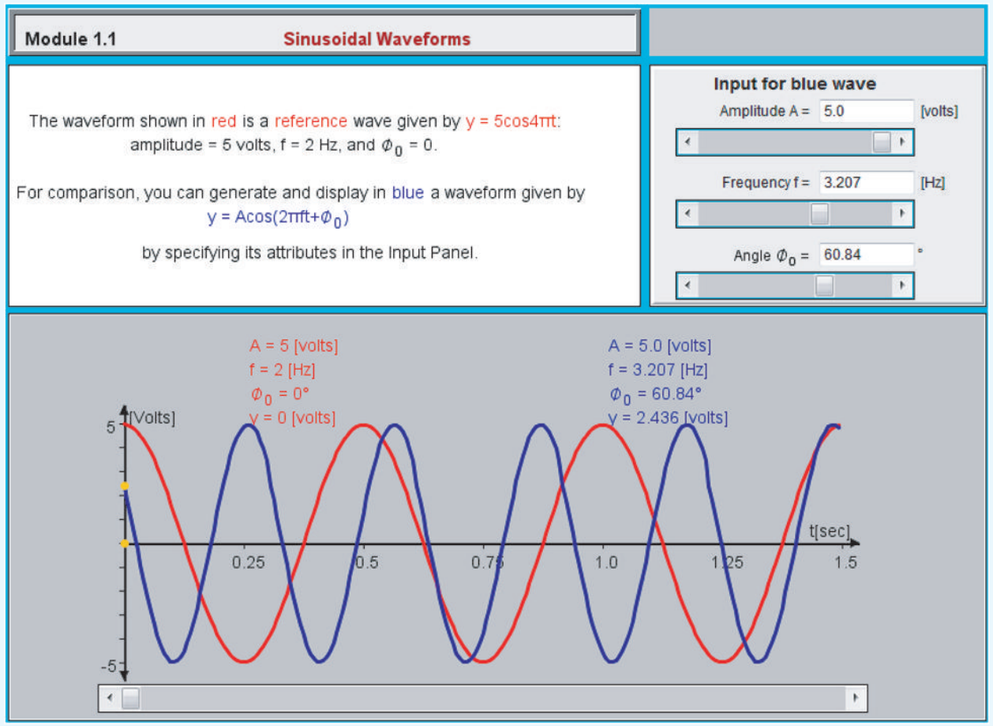
(4) $v = 5\sin 2\pi t/8$,



图E1.4 横轴t（s），纵轴电压v（V）。

答案： （a）第（2）式，超前参考余弦波$\pi/4$；（b）第（4）式，$5\sin(2\pi t/8)$。详解参见原书配套EM资源。

Module 1.1 Sinusoidal Waveforms Learn how the shape of the waveform is related to the amplitude, frequency, and reference phase angle of a sinusoidal wave.



Exercise 1-5: The electric field of a traveling electromagnetic wave is given by

$$E(z,t) = 10\cos(\pi \times 10^7 t + \pi z/15 + \pi/6) \qquad \text{(V/m)}.$$

Determine (a) the direction of wave propagation, (b) the wave frequency f, (c) its wavelength λ, and (d) its phase velocity u_p.

Answer: (a) $-z$ direction, (b) $f = 5$ MHz, (c) $\lambda = 30$ m, (d) $u_p = 1.5 \times 10^8$ m/s. (See EM.)

1-4.2 Sinusoidal Waves in a Lossy Medium

If a wave is traveling in the x direction in a ***lossy medium***, its amplitude decreases as $e^{-\alpha x}$. This factor is called the ***attenuation factor***, and α is called the ***attenuation constant*** of the medium and its unit is neper per meter (Np/m). Thus, in general,

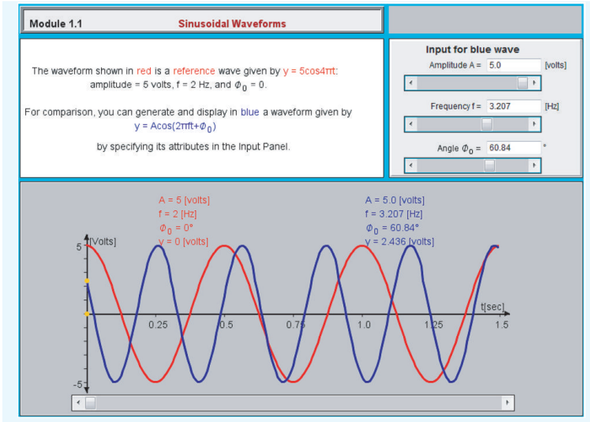
$$y(x,t) = Ae^{-\alpha x}\cos(\omega t - \beta x + \phi_0). \qquad (1.33)$$

The wave amplitude is now $Ae^{-\alpha x}$, not just A. **Figure 1-15** shows a plot of $y(x,t)$ as a function of x at $t = 0$ for $A = 10$ m, $\lambda = 2$ m, $\alpha = 0.2$ Np/m, and $\phi_0 = 0$. Note that the envelope of the wave pattern decreases as $e^{-\alpha x}$.

正弦波练习与有损介质

模块1.1 正弦波形

通过配套交互模块，观察正弦波的振幅、频率和参考相位角如何影响波形。下图保留原书模块界面。



原书模块1.1：参数调节与正弦波形显示。

练习1-5

某行进电磁波的电场为：

$$E(z,t) = 10\cos(\pi \times 10^7 t + \pi z/15 + \pi/6) \qquad \text{(V/m)}.$$

求（a）传播方向；（b）频率f；（c）波长λ；（d）相速度u_p。

答案： t、z两项同号，波沿−z方向传播；$\omega = \pi \times 10^7$ rad/s，故$f = \omega/(2\pi) = 5$ MHz；$\beta = \pi/15$ rad/m，故$\lambda = 2\pi/\beta = 30$ m；$u_p = f\lambda = 1.5 \times 10^8$ m/s。

1-4.2 有损介质中的正弦波

波在有损介质中沿+x方向传播时，振幅随传播距离按因子$e^{-\alpha x}$衰减。$e^{-\alpha x}$称为衰减因子，α为衰减常数，单位是奈培每米（Np/m）。因此一般可写为：

$$y(x,t) = Ae^{-\alpha x}\cos(\omega t - \beta x + \phi_0). \qquad (1.33)$$

此时振幅为$Ae^{-\alpha x}$，而非A，更不能把振幅直接写成无量纲的$e^{-\alpha x}$。图1-15给出A = 10 m、λ = 2 m、α = 0.2 Np/m、$\varphi_0 = 0$时，在t = 0的波形；其正、负包络分别为$\pm 10e^{-0.2x}$。

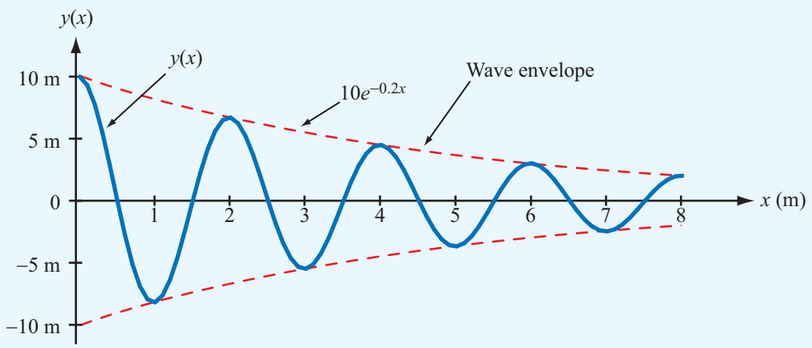


Figure 1-15 Plot of $y(x) = (10e^{-0.2x}\cos \pi x)$ meters. Note that the envelope is bounded between the curve given by $10e^{-0.2x}$ and its mirror image.

The real unit of α is (1/m); the neper (Np) part is a dimensionless, artificial adjective traditionally used as a reminder that the unit (Np/m) refers to the attenuation constant of the medium, α. A similar practice is applied to the phase constant β by assigning it the unit (rad/m) instead of just (l/m).

Concept Question 1-6: How can you tell if a wave is traveling in the positive x direction or the negative x direction?

Concept Question 1-7: How does the envelope of the wave pattern vary with distance in (a) a lossless medium and (b) a lossy medium?

Concept Question 1-8: Why does a negative value of ϕ_0 signify a phase lag?

Example 1-1: Sound Wave in Water

An acoustic wave traveling in the x direction in a fluid (liquid or gas) is characterized by a differential pressure $p(x,t)$. The unit for pressure is newton per square meter (N/m^2). Find an expression for $p(x,t)$ for a sinusoidal sound wave traveling in the positive x direction in water given that the wave frequency is 1 kHz, the velocity of sound in water is 1.5 km/s, the wave amplitude is 10 N/m^2, and $p(x,t)$ was observed to be at its maximum value at $t = 0$ and $x = 0.25$ m. Treat water as a lossless medium.

Solution: According to the general form given by Eq. (1.17) for a wave traveling in the positive x direction,

$$p(x,t) = A\cos\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x + \phi_0\right) \quad (\text{N/m}^2).$$

The amplitude $A = 10$ N/m^2, $T = 1/f = 10^{-3}$ s, and from $u_p = f\lambda$,

$$\lambda = \frac{u_p}{f} = \frac{1.5\times 10^3}{10^3} = 1.5 \text{ m}.$$

Hence,

$$p(x,t) = 10\cos\left(2\pi\times 10^3 t - \frac{4\pi}{3}x + \phi_0\right) \quad (\text{N/m}^2).$$

Since at $t = 0$ and $x = 0.25$ m, $p(0.25,0) = 10$ N/m^2, we have

$$10 = 10\cos\left(\frac{-4\pi}{3}0.25 + \phi_0\right) = 10\cos\left(\frac{-\pi}{3} + \phi_0\right),$$

which yields the result $(\phi_0 - \pi/3) = \cos^{-1}(1)$, or $\phi_0 = \pi/3$. Hence,

$$p(x,t) = 10\cos\left(2\pi\times 10^3 t - \frac{4\pi}{3}x + \frac{\pi}{3}\right) \quad (\text{N/m}^2).$$

衰减包络与水中声波示例

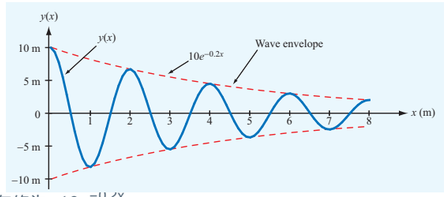


图1-15 $y(x) = 10e^{-0.2x}\cos(\pi x)$ m；上下包络为$\pm 10e^{-0.2x}$ m。

从量纲看，α的单位为m^{-1}。奈培（Np）用于标明其对应以自然对数定义的衰减量；Np/m强调这是衰减常数。类似地，β采用rad/m，以区分单位长度的相位变化与一般的倒长度量。

概念题1-6：如何判断波沿+x还是−x方向传播？
概念题1-7：排除几何扩散后，波的包络在（a）无损介质；（b）有损介质中如何随距离变化？
概念题1-8：为什么在本书时间约定下，负φ_0表示相位滞后？

例1-1 水中的声波

流体中的声波可用相对静压的声压扰动p(x,t)描述，单位为帕斯卡，即N/m^2。已知水中正弦声波沿+x方向传播，频率1 kHz，声速1.5 km/s，声压振幅10 N/m^2，且在t = 0、x = 0.25 m处达到正最大值。将水视为无损介质，求p(x,t)。

解：按式（1.17）的形式：

$$p(x,t) = A\cos\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x + \phi_0\right) \quad (\text{N/m}^2).$$

A = 10 N/m^2，T = 1/f = 10^{-3} s。由$u_p = f\lambda$得：

$$\lambda = \frac{u_p}{f} = \frac{1.5\times 10^3}{10^3} = 1.5 \text{ m}.$$

代入周期和波长：

$$p(x,t) = 10\cos\left(2\pi\times 10^3 t - \frac{4\pi}{3}x + \phi_0\right) \quad (\text{N/m}^2).$$

将p(0.25,0) = 10 N/m^2代入，有：

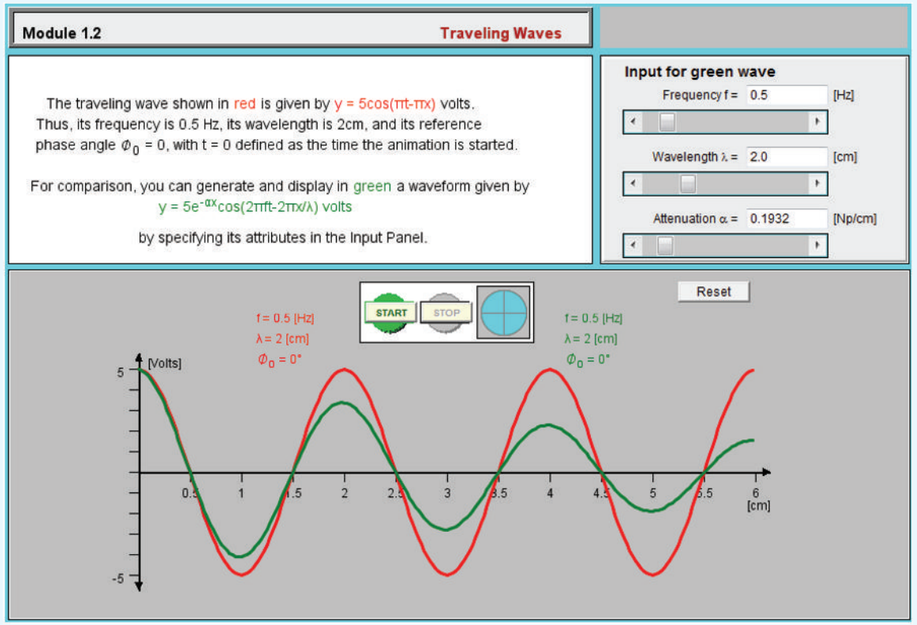
$$10 = 10\cos\left(\frac{-4\pi}{3}0.25 + \phi_0\right) = 10\cos\left(\frac{-\pi}{3} + \phi_0\right),$$

因此$\varphi_0 - \pi/3 = 2n\pi$；取代表值$\varphi_0 = \pi/3$，最终得到：

$$p(x,t) = 10\cos\left(2\pi\times 10^3 t - \frac{4\pi}{3}x + \frac{\pi}{3}\right) \quad (\text{N/m}^2).$$

核算：λ = 1500/1000 = 1.5 m，β = 2π/λ = 4π/3 rad/m；x = 0.25 m、t = 0时相位为零，声压确为+10 N/m^2。原书给出的结果通过代入检查。

Module 1.2 Traveling Waves Learn how the shape of a traveling wave is related to its frequency and wavelength and to the attenuation constant of the medium.



Example 1-2: Power Loss

A laser beam of light propagating through the atmosphere is characterized by an electric field given by

$$E(x,t) = 150e^{-0.03x}\cos(3\times10^{15}t - 10^{7}x) \qquad \text{(V/m)},$$

where x is the distance from the source in meters. The attenuation is due to absorption by atmospheric gases. Determine

(a) the direction of wave travel,

(b) the wave velocity, and

(c) the wave amplitude at a distance of 200 m.

Solution: (a) Since the coefficients of t and x in the argument of the cosine function have opposite signs, the wave must be traveling in the $+x$ direction.

(b)

$$u_p = \frac{\omega}{\beta} = \frac{3\times10^{15}}{10^{7}} = 3\times10^{8} \text{ m/s},$$

and this is equal to c, which is the velocity of light in free space.

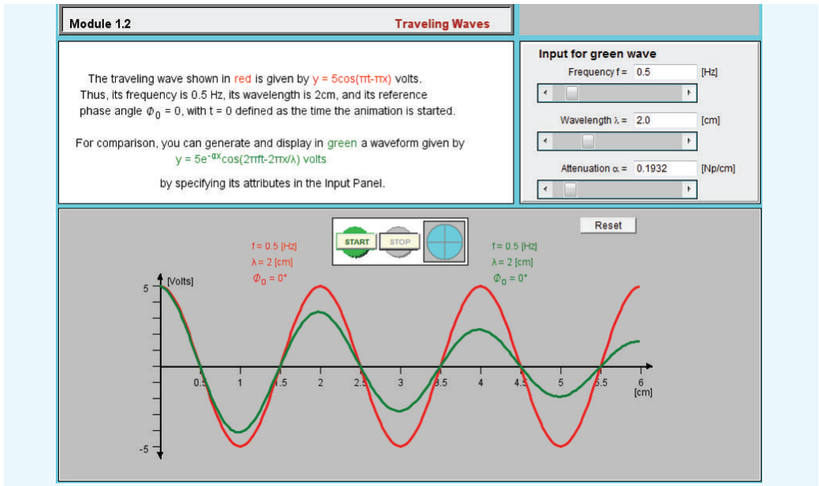
(c) At $x = 200$ m, the amplitude of $E(x,t)$ is

$$150e^{-0.03\times200} = 0.37 \qquad \text{(V/m)}.$$

Exercise 1-6: Consider the red wave shown in **Fig. E1.6**. What is the wave's (a) amplitude (at $x = 0$), (b) wavelength, and (c) attenuation constant?

模块1.2与例1-2：功率损耗

模块1.2 行波：观察行波的频率、波长以及介质衰减常数对波形的影响。



原书模块1.2：行波传播、相位与衰减。

例1-2 功率损耗

一束激光在大气中传播，其电场为：

$$E(x,t) = 150e^{-0.03x}\cos(3\times10^{15}t - 10^{7}x) \qquad \text{(V/m)},$$

x是距光源的距离，单位为米。大气气体吸收导致衰减。求（a）波的传播方向；（b）波速；（c）距光源200 m处的电场振幅。

解：（a）余弦相位中t项与x项异号，因此沿+x方向传播。

$$u_p = \frac{\omega}{\beta} = \frac{3\times10^{15}}{10^{7}} = 3\times10^{8} \text{ m/s},$$

（b）上述相速度约等于真空光速c。
（c）x = 200 m处的电场振幅为：

$$150e^{-0.03\times200} = 0.37 \qquad \text{(V/m)}.$$

数值复核：$150e^{-6} = 0.3718$ V/m，按原书精度为0.37 V/m。Power loss应译为“功率损耗”，不能译为“断电”；本例具体所求为场的振幅。

练习1-6

观察下页图E1.6的红色波形，求（a）x = 0处振幅；（b）波长；（c）衰减常数。

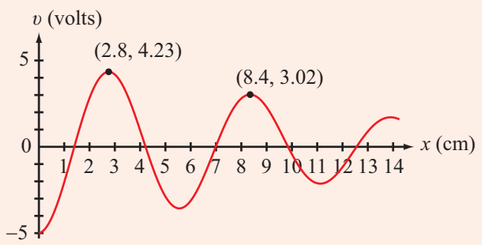


Figure E1.6

Answer: (a) 5 V, (b) 5.6 cm, (c) $\alpha = 0.06$ Np/cm. (See EM.)

Exercise 1-7: The red wave shown in **Fig. E1.7** is given by $\upsilon = 5\cos 4\pi x$ (V). What expression is applicable to (a) the blue wave and (b) the green wave?

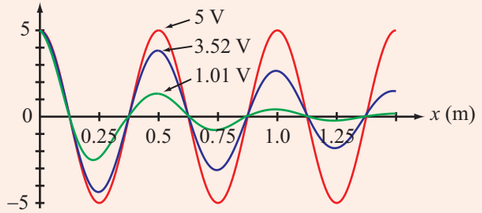


Figure E1.7

Answer: (a) $\upsilon = 5e^{-0.7x}\cos 4\pi x$ (V),
(b) $\upsilon = 5e^{-3.2x}\cos 4\pi x$ (V). (See EM.)

Exercise 1-8: An electromagnetic wave is propagating in the z direction in a lossy medium with attenuation constant $\alpha = 0.5$ Np/m. If the wave's electric-field amplitude is 100 V/m at $z = 0$, how far can the wave travel before its amplitude is reduced to (a) 10 V/m, (b) 1 V/m, and (c) 1 μV/m?

Answer: (a) 4.6 m, (b) 9.2 m, (c) 37 m. (See EM.)

1-5 The Electromagnetic Spectrum

Visible light belongs to a family of waves arranged according to frequency and wavelength along a continuum called the ***electromagnetic spectrum*** (**Fig. 1-16**). Other members of this family include gamma rays, X-rays, infrared waves, and radio waves. Generically, they all are called EM waves because they share the following fundamental properties:

- A ***monochromatic*** (single frequency) EM wave consists of electric and magnetic fields that oscillate at the same frequency f.
- The phase velocity of an EM wave propagating in a vacuum is a universal constant given by the velocity of light c, defined earlier by Eq. (1.14).
- In a vacuum, the wavelength λ of an EM wave is related to its oscillation frequency f by

$$\lambda = \frac{c}{f}. \tag{1.34}$$

Whereas all monochromatic EM waves share these properties, each is distinguished by its own wavelength λ, or equivalently by its own oscillation frequency f.

The visible part of the EM spectrum shown in **Fig. 1-16** covers a very narrow wavelength range extending between $\lambda = 0.4$ μm (violet) and $\lambda = 0.7$ μm (red). As we move progressively toward shorter wavelengths, we encounter the ultraviolet, X-ray, and gamma-ray bands, each so named because of historical reasons associated with the discovery of waves with those wavelengths. On the other side of the visible spectrum lie the infrared band and then the microwave part of the radio region. Because of the link between λ and f given by Eq. (1.34), each of these spectral ranges may be specified in terms of its wavelength range or its frequency range. In practice, however, a wave is specified in terms of its wavelength λ if $\lambda < 1$ mm, which encompasses all parts of the EM spectrum except for the radio region, and the wave is specified in terms of its frequency f if $\lambda > 1$ mm (i.e., in the radio region). A wavelength of 1 mm corresponds to a frequency of 3×10^{11} Hz = 300 GHz in free space.

The radio spectrum consists of several individual bands, as shown in 'the chart of **Fig. 1-17** (see p. 32). Each band covers one decade of the radio spectrum and has a letter designation based on a nomenclature defined by the International Telecommunication Union. Waves of different frequencies have different applications because they are excited by different mechanisms, and the properties of an EM wave propagating in a nonvacuum material may vary considerably from one band to another.

Although no precise definition exists for the extent of the ***microwave band***, it is conventionally regarded to cover the full ranges of the UHF, SHF, and EHF bands. The EHF band is sometimes referred to as the ***millimeter-wave band*** because

衰减练习与1-5电磁波谱

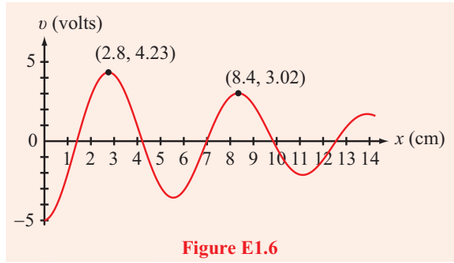


Figure E1.6

图E1.6，横轴x（cm），纵轴电压v（V）。

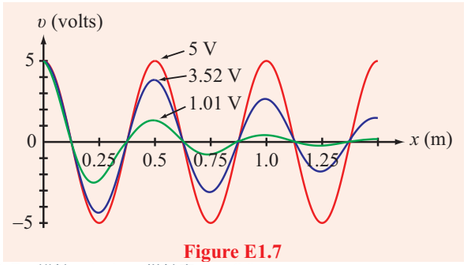


Figure E1.7

图E1.7，横轴x（m），纵轴电压v（V）。

练习1-6答案： （a）5 V；（b）λ = 5.6 cm；（c）由两个包络采样值，α = ln(4.23/3.02)/(8.4−2.8) ≈ 0.060 Np/cm。

练习1-7： 图E1.7红色波形为v = 5 cos(4πx) V，求蓝色、绿色波形的表达式。
答案： 蓝色：v = $5e^{-0.7x}$ cos(4πx) V；绿色：v = $5e^{-3.2x}$ cos(4πx) V，其中x以米计。

练习1-8： 电磁波沿+z方向在有损介质中传播，α = 0.5 Np/m，z = 0处电场振幅为100 V/m。振幅分别降到10 V/m、1 V/m和1 μV/m时，传播距离是多少？
答案： 由z = ln(100/A)/0.5得到约4.6 m、9.2 m和37 m。第三问1 μV/m = 10^{-6} V/m。

1-5 电磁波谱

按频率或波长连续排列的电磁波构成电磁波谱（图1-16），包括伽马射线、X射线、紫外线、可见光、红外线和无线电波。它们具有以下共同性质：单色波的电场与磁场以同一频率f振荡；在真空中其相速度为式（1.14）的c；真空波长满足：

$$\lambda = \frac{c}{f}. \tag{1.34}$$

虽然具有共同性质，各单色波仍可由各自的频率或真空波长区分。原书给出可见光范围约0.4 μm（紫）至0.7 μm（红）。更短波长依次属于紫外、X射线和伽马射线；更长波长则进入红外与无线电波段。这些名称与发现历史有关。

任何波段均可用频率或波长表示。原书采用的惯例是：波长小于1 mm时常用波长，大于1 mm时常用频率。真空中1 mm对应300 GHz。

图1-17将无线电波谱分为若干频段。各标准频段覆盖一个**十倍频程**（上下限频率相差10倍），其字母名称采用国际电信联盟的命名。不同频段的产生机制、介质传播性质及应用有所不同。

微波频段没有唯一边界定义；原书采用UHF、SHF、EHF三个频段合计的范围。EHF也称毫米波段，波长为1 mm至1 cm，对应300 GHz至30 GHz。

核算：练习1-8三个距离的未舍入值分别约4.605、9.210、36.841 m。原书频段划分图属于历史资料。